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Pile Strength Variation of a Bridge Abutment: A Case Study of SASEC Dhaka Sylhet Corridor Road Investment Project

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ABSTRACT

Pile plays an important role in the foundation of bridge construction. The main purpose of piles in bridge construction is to transfer the structure's load from a low-bearing capacity soil layer to a high-bearing capacity soil layer. In this study, we have selected the Islamabad bridge as part of the SASEC Dhaka Sylhet Corridor Road Investment Project, package No. DS05. The bridge has two abutments: A1 and A2. Each abutment is supported by 30 piles that measure 24 meters in length and 1.2 meters in diameter. As all the piles were not cast together, some strength variation may occur from pile to pile. This study finds that, after 7,14 & 28 days, the strength variation of piles in the A1 abutment is around 3 to 28.2%, 3.8 to 26.2%, and 3 to 25.35%. Variations in the strength of piles may be attributed to the factors such as temperature, road traffic, and the distance between the batching plant and the construction site. This study also finds that, after 7 days, maximum piles have achieved their target strength, and after 28 days, they became almost double their target strength. This may have occurred due to the safety factors considered in the concrete mix design.

Keywords: *SASEC, strength, pile, abutment, concrete*

INTRODUCTION

Bridges are important parts of infrastructure and play a crucial role in developing the economic growth of countries [1]. Bridges are vital components of transportation infrastructure networks since they facilitate economic growth and provide a means for vehicles and pedestrians to cross obstacles like rivers and valleys [2]. The most significant part of a bridge's strength (stability) is its substructure(base), which is strengthened by piles. Piles are the core members of the bridge foundation. This study looks closely at how various factors (Temperature, Transportation Time, and Weather Conditions) are affecting the properties of piles for a single-span bridge (BR-21) in the SASEC Dhaka Sylhet Corridor Road Investment Project. This bridge has two abutments (A1 & A2), and each end has 30 piles supporting it. The most challenging part is to cast all 30 piles at the same day because of logistical problems. As a consequence, they have to be casted at different times and on different days, which indicates a slight variation in the concrete properties.

For the purpose of determining the overall quality of concrete as well as its resistance, compressive strength is utilized [3]. Concrete's compressive strength is typically measured by subjecting a hardened specimen to a gradually increasing axial compression load [4]. Nonetheless, aside from durability, compressive strength is one of the most fundamental specialized features of concrete. [5,6]. This investigation entails a comprehensive examination of concrete samples from each pile, subjected to rigorous strength testing at crucial intervals of 7, 14, and 28 days. The ensuing analysis illuminates a substantial 3 to 26% disparity in pile strength within a single abutment. This variability is attributed to a confluence of factors, including the distance between the batching plant and the construction site, temperature fluctuations, and disruptions caused by traffic delays. In construction projects, the strength of concrete is a crucial aspect that can be affected by multiple external factors. Temperature is particularly significant, as it influences the hydration process, ultimately impacting the structural integrity of the concrete. Moreover, the distance between the batching plant and the construction site can affect concrete quality due to transportation factors and time delays. Furthermore, variations in concrete strength can arise from traffic conditions during transit. A comprehensive understanding of these interconnected factors is vital for guaranteeing the durability and performance of concrete structures across various environmental and logistical settings. In addition, the compressive strength of concrete is influenced by the proportions of its mixture, which include the weight-to-cement ratio (w/c), cement content, the ratio of fine to coarse aggregate (FA/CA), and the total volume of aggregate [7].

METHODOLOGY

In this study, we have selected the proposed Islamabad bridge as part of the SASEC Dhaka Sylhet Corridor Road Investment Project, package No. DS05. The proposed bridge has two abutments: A1 and A2. Each abutment is supported by 30 piles that measure 24 meters in length and 1.2 meters in diameter. Only the A1 abutment's piles are being considered for this study. Concrete samples were collected for each pile and tested for their compressive strength after 7, 14, and 28 days. Further information is provided below.

Table 1: *Details of concrete specification.*

Concrete Grade	C25
Slump Value	Between 160 to 200 mm
Cement Brand	Seven Rings (OPC)
Admixture Brand	Sikaplast 2028 NS
Distance Between Batching Plant to Construction Site	11.2 km

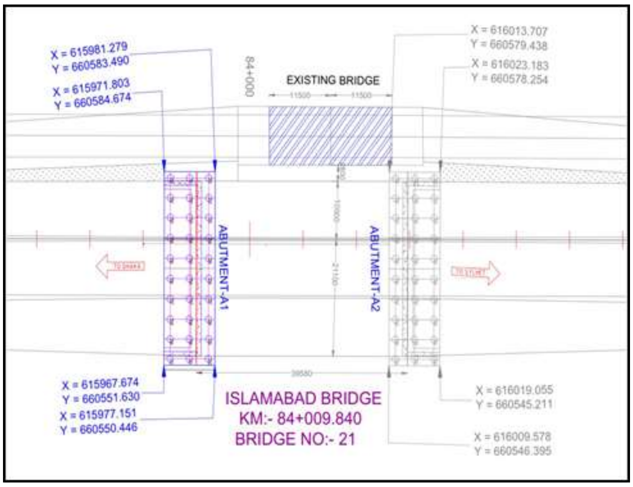

Fig. 1: *Plan view of abutment A1 and A2.*

Fig. 2: *Concrete sample collection from site of Islamabad bridge.*

Fig. 3: *Curing of sample cylinders.*

Fig. 4: *Compressive Strength test of sample Cylinders.*

Fig. 5: Road distance between batching plant and construction site.

RESULTS AND DISCUSSION

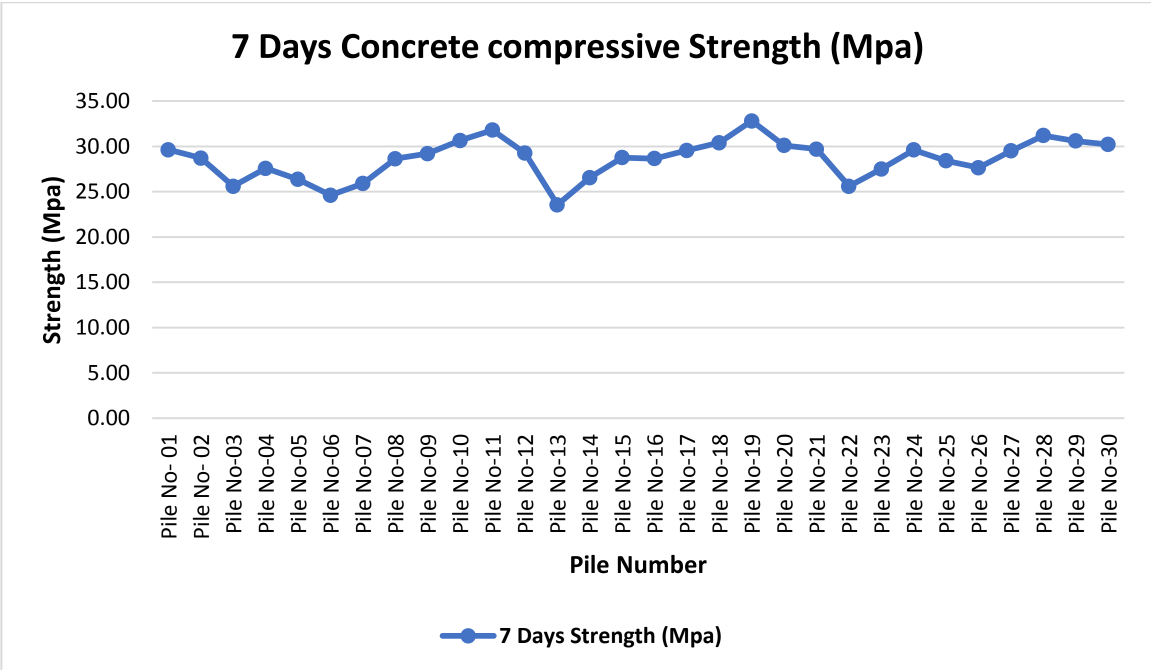

***Chart 1:** Concrete compressive strength of piles on 7th day (Abutment A1).*

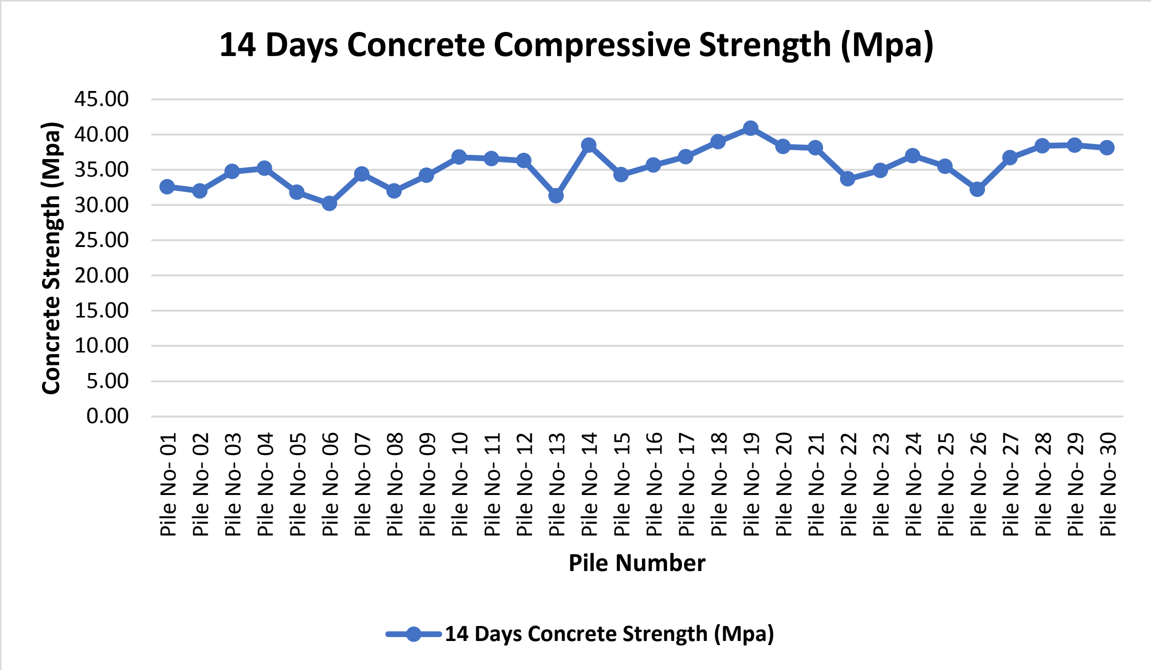

Chart 2: *Concrete compressive strength of piles on 14th day (Abutment A1).*

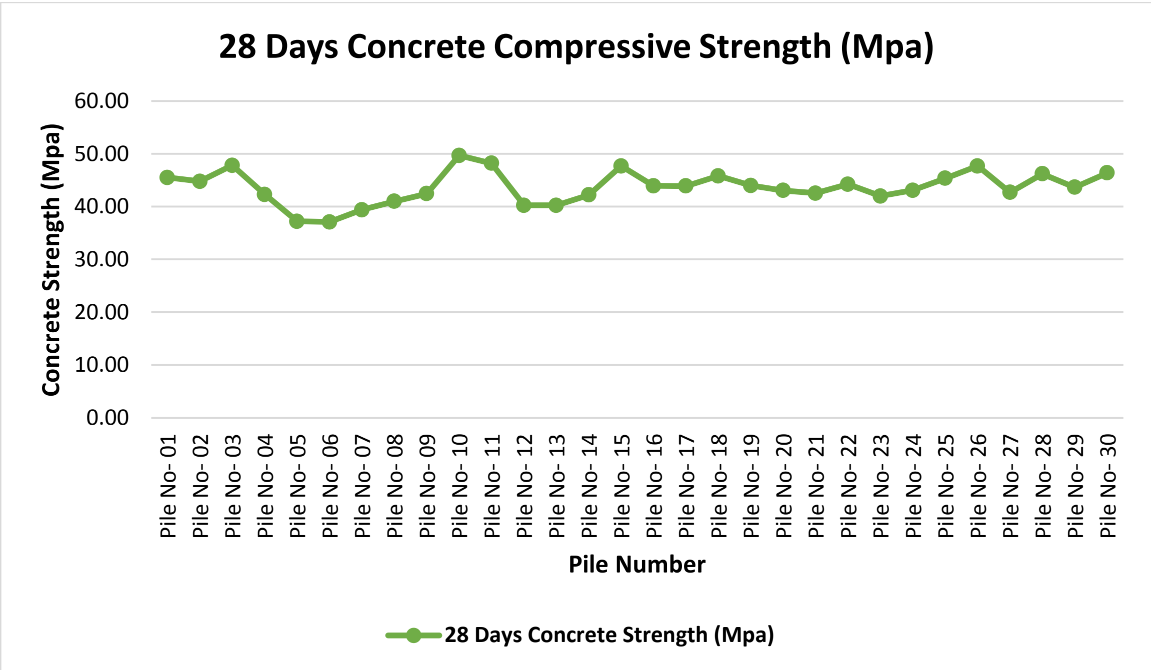

Chart 3: *Concrete compressive strength of piles on 28th day (Abutment A1).*

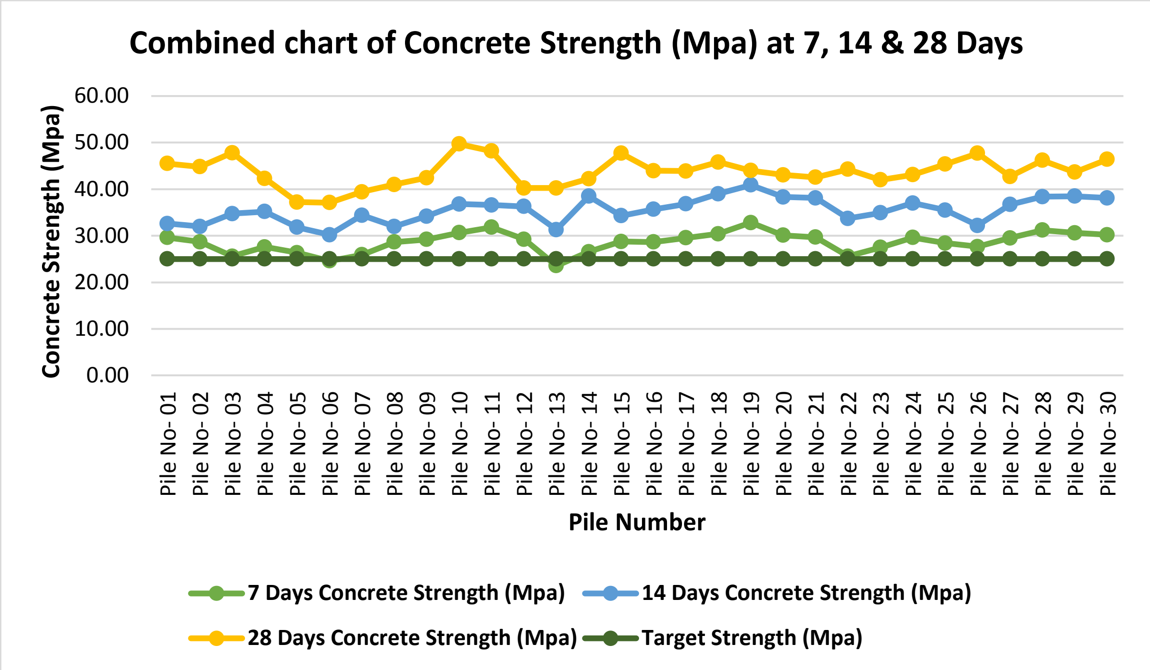

Chart 4: *Comparison of Concrete compressive strength of piles (Abutment A1).*

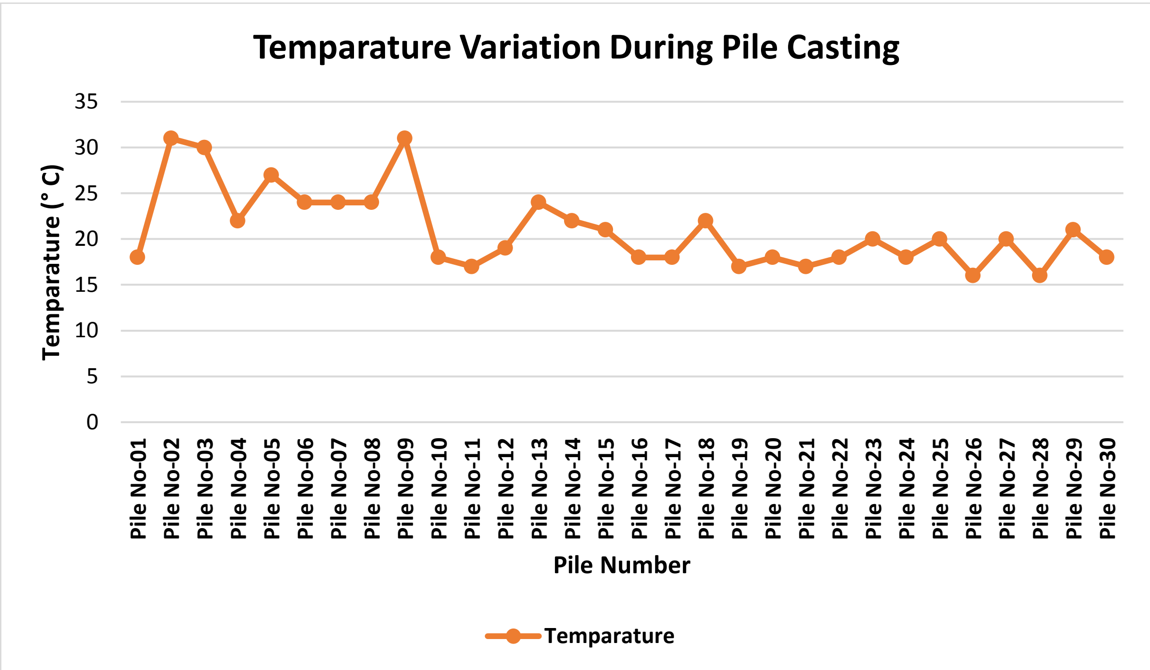

Chart 5: *Temperature variation during pile casting.*

Table 2: *Summery of compressive strength of piles at 7,14 & 28 days and temperature during Pile casting.*

Pile Number	7 Days Compressive Strength (Mpa)	14 Days Compressive Strength (Mpa)	28 Days Compressive Strength (Mpa)	Temperature While pile casting (°C)	Pile Number	7 Days Compressive Strength (Mpa)	14 Days Compressive Strength (Mpa)	28 Days Compressive Strength (Mpa)	Temperature While pile casting (°C)
01	29.63	32.60	45.50	18	16	28.65	35.66	43.95	18
02	28.70	32.00	44.80	31	17	29.55	36.85	43.90	18
03	25.60	34.74	47.80	30	18	30.40	39.00	45.80	22
04	27.58	35.20	42.30	22	19	32.80	40.90	44.00	17
05	26.35	31.80	37.20	27	20	30.10	38.30	43.05	18
06	24.60	30.20	37.10	24	21	29.70	38.10	42.55	17
07	25.90	34.40	39.40	24	22	25.60	33.70	44.25	18
08	28.64	32.00	41.00	24	23	27.50	34.90	42.00	20
09	29.20	34.20	42.45	31	24	29.60	37.00	43.10	18
10	30.65	36.80	49.70	18	25	28.40	35.50	45.35	20
11	31.80	36.59	48.20	17	26	27.65	32.20	47.70	16
12	29.25	36.30	40.25	19	27	29.50	36.70	42.70	20
13	23.55	31.30	40.25	24	28	31.20	38.40	46.20	16
14	26.55	38.50	42.22	22	29	30.60	38.50	43.65	21
15	28.75	34.30	47.70	21	30	30.20	38.10	46.40	18

Charts 1,2,3 show the concrete compressive strength of piles on the 7^{th},$14^{th,}$ and 28^{th} day of bridge Abutment A1. These three charts reveal that within 7 days around 94.3% of piles have reached the targeted strength, on the 14^{th} day (shown in chart 2), the Concrete compressive strength was 1.2 to 1.5 times the targeted strength and on the 28^{th} day (shown in chart 2), the piles reached the strength in between 40 to 50MPa which is two times of the targeted strength of 25 MPa.

This might have happened due to the safety factors considered in the concrete mix design. Charts 1,2,3 also show the variation of compressive strength of piles on the 7^{th},$14^{th,}$ and 28^{th} day of bridge Abutment A1. After 7 days the strength variation of piles was around 3.0 to 28.2%, after 14 days it was 3.0 to 26.2% and after 28 days the variation was 3.0 to 25.35%.

The variation may accrue as all the piles are cast on different days and times, and the temperature and field conditions are not uniform. Pile wise temperature variation is shown in chat 05. Also, the distance between the batching plant and the construction site (shown in Figure 05) is responsible for the variation. As the concrete mixer truck was transported by road, the transportation period duration depended on the road traffic situation. So, due to the surrounding facts, the pile strength variation happened.

CONCLUSION

- This study finds that, an abutment may have piles of different strengths and pile to pile strength variations can be 28% or more.
- Maximum strength variation was found 28.2% on 7th days. Variations in the strength of piles may be attributed to the factors such as temperature, road traffic, and the distance between the batching plant and the construction site.
- The strength variation in piles will not impact the integrity of the structure as maximum piles have achieved almost double of their target strength after 28 days.
- As per this study, if the variation of strength in piles exceeds 50%, it will be alarming for these structures.

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